COLLEGE MANAGEMENT SYSTEM USING FILE OPERATIONS IN PYTHON

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SUBJECT : PYTHON

DEPARTMENT : CSE(CYBER SECUIRITY)

BATCH : 03

1 AIM

To develop a College Management System using Python file operations (CSV) to add, view, search, and delete student records efficiently.

2 PROCEDURE

Step 1: Start the program.

Step 2: Import csv and os modules.

Step 3: Check if the file exists, otherwise create it with headers.

Step 4: Define function to add student details into the file.

Step 5: Define function to view all student records.

Step 6: Define function to search student by ID.

Step 7: Define function to delete student record using ID.

Step 8: Handle exceptions for invalid inputs.

Step 9: Execute all functions sequentially.

Step 10: Display final updated student list and stop the program.

3 CODE

```
import csv
import os
```

```
FILE = "students.csv"

if not os.path.exists(FILE):
    with open(FILE, "w", newline="") as f:
        writer = csv.writer(f)
        writer.writerow(["ID", "Name", "Age", "Course"])

def add_student():
    try:
        sid = input("Enter ID: ")
        name = input("Enter Name: ")
        age = int(input("Enter Age: "))
        course = input("Enter Course: ")

        with open(FILE, "a", newline="") as f:
            writer = csv.writer(f)
            writer.writerow([sid, name, age, course])

        print("Student added successfully!")

    except Exception as e:
        print("Error:", e)

def view_students():
    with open(FILE, "r") as f:
        reader = csv.reader(f)
        for row in reader:
            print(row)

def search_student():
    sid = input("Enter ID to search: ")
    found = False

    with open(FILE, "r") as f:
        reader = csv.reader(f)
        for row in reader:
            if row[0] == sid:
                print("Found:", row)
                found = True
                break

    if not found:
        print("Student not found!")

def delete_student():
```

```
    sid = input("Enter ID to delete: ")
    rows = []

    with open(FILE, "r") as f:
        reader = csv.reader(f)
        rows = list(reader)

    found = False

    with open(FILE, "w", newline="") as f:
        writer = csv.writer(f)
        for row in rows:
            if row[0] != sid:
                writer.writerow(row)
            else:
                found = True

    if found:
        print("Student deleted!")
    else:
        print("Student not found!")

print("---- Add Student ----")
add_student()

print("\n---- View Students ----")
view_students()

print("\n---- Search Student ----")
search_student()

print("\n---- Delete Student ----")
delete_student()

print("\n---- Final List ----")
view_students()
```

4 OUTPUT

```
---- Add Student ----
Enter ID: 101
Enter Name: John
Enter Age: 20
Enter Course: CSE
```

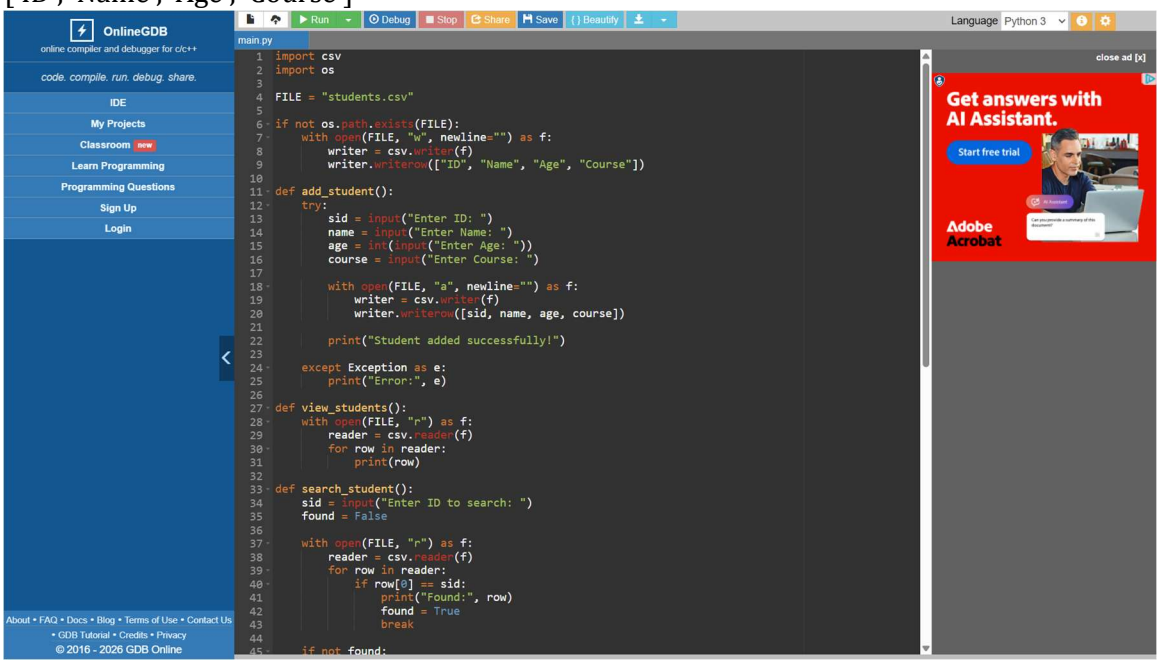
Student added successfully!

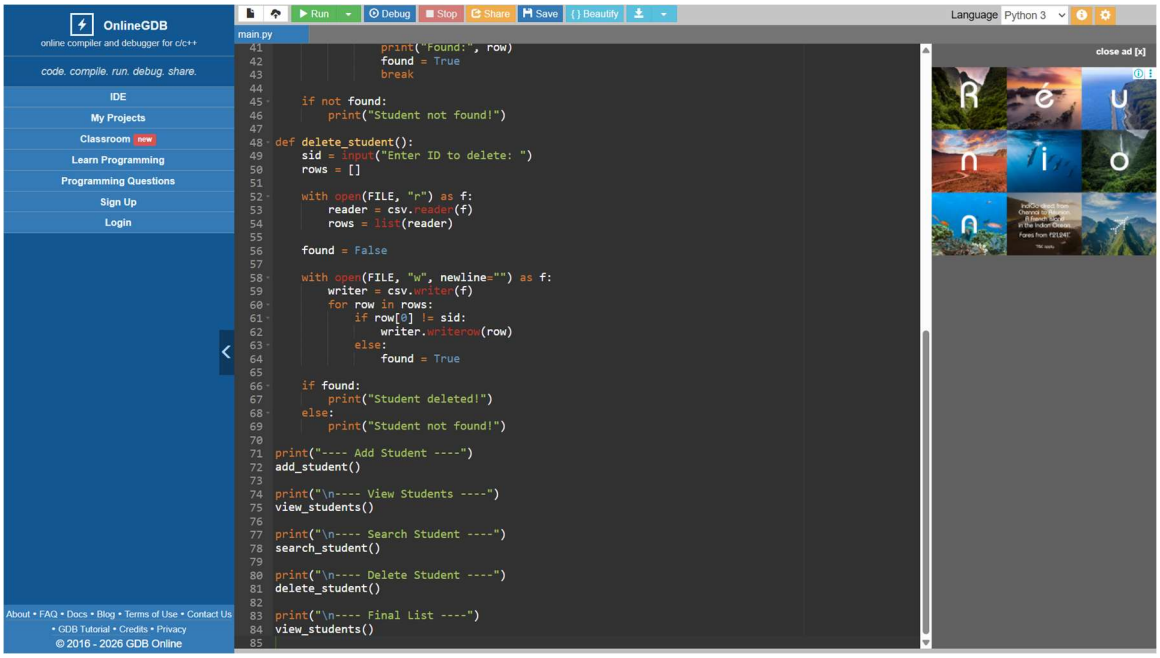
---- View Students ----
['ID', 'Name', 'Age', 'Course']
['101', 'John', '20', 'CSE']

---- Search Student ----
Enter ID to search: 101
Found: ['101', 'John', '20', 'CSE']

---- Delete Student ----
Enter ID to delete: 101
Student deleted!

---- Final List ----
['ID', 'Name', 'Age', 'Course']






```python
                print("Found:", row)
                found = True
                break

    if not found:
        print("Student not found!")

def delete_student():
    sid = input("Enter ID to delete: ")
    rows = []

    with open(FILE, "r") as f:
        reader = csv.reader(f)
        rows = list(reader)

    found = False

    with open(FILE, "w", newline="") as f:
        writer = csv.writer(f)
        for row in rows:
            if row[0] != sid:
                writer.writerow(row)
            else:
                found = True

    if found:
        print("Student deleted!")
    else:
        print("Student not found!")

print("---- Add Student ----")
add_student()

print("\n---- View Students ----")
view_students()

print("\n---- Search Student ----")
search_student()

print("\n---- Delete Student ----")
delete_student()

print("\n---- Final List ----")
view_students()
```

5 RESULT

Thus, the College Management System using file operations (CSV) in Python has been successfully implemented with add, view, search, and delete functionalities.